

Xebia and dsm-firmenich Help Farmers Meet Their Sustainability Targets on AWS with Sustell

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“By building on AWS, rather than a traditional infrastructure, we can increase in-house skills and keep a tight control on costs.” — *Andres Koetsier, Cloud Native Development Practice Lead, Xebia*

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Exploring Sustainable Solutions for Animal Farmers

As demand for sustainable products grows, animal farmers are looking for ways to reduce their environmental footprint, which is not possible without visibility into their day-to-day activities. Data on water usage, carbon emissions, and other factors can help animal farmers make decisions to help achieve their sustainability goals, but many don't have the capability to prepare or review a comprehensive view of their

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operations. To overcome this challenge, [dsm-firmenich](#) has created Sustell, an intelligent solution for sustainability-related applications built on Amazon Web Services (AWS).

Built with the help of AWS Premier Tier Services Partner [Xebia](#), Sustell makes the invisible visible, showing users data to help them understand their environmental impact and identify best practices across multiple facets of their business. With this solution, animal farmers can improve operational efficiency, access sustainable financing options, and reach increasingly stringent environmental targets.

Powering Sustell with Xebia and AWS Technologies

The company has group-wide ambitions to limit its environmental impact and help its customers do the same. With a division focused on animal nutrition, dsm-firmenich knows that animal farmers (and the animal protein value chain) often struggle to assess their carbon footprint, which can affect their ability to meet their sustainability goals and their respective consumer requirements.

“Consumers are willing to pay more for credibly measured and certified lower-carbon food,” says David Nickell, vice president of sustainability and business solutions at dsm-firmenich Animal Nutrition & Health. In fact, a joint study from [McKinsey and NielsenIQ](#) found that 78 percent of US consumers value a sustainable lifestyle.

“Animal farmers can’t improve what they don’t measure,” says Nickell. “But they also need ways to report their environmental impact to licensing bodies, regulators, their customers, and banks to operate and apply for subsidies.”

dsm-firmenich built Sustell, a scalable, user-friendly solution that tracks data from 19 separate categories to help users visualize their environmental impact. These include the consumption of fresh water; emissions from fossil fuels and peat contributing to elevated greenhouse gas levels in the atmosphere; the depletion of hydrocarbon-rich resources such as coal, oil, and natural gas; and photochemical smog, which consists of carbon monoxide, sulfur oxides, smoke, and other pollutants. This solution, certified by the International Organization for Standardization, empowers feed producers and animal farmers to assess the environmental impact of their business at scale and speed.

Xebia, an IT consultancy and software development company, provided critical support throughout the process. This AWS Partner was involved in every stage of the project, from designing initial mock-ups and mapping the user story to implementing the application.



Generating Environmental Data for Insights and Audits

Xebia and dsm-firmenich run Sustell on a serverless AWS infrastructure powered by solutions such as [AWS Lambda](#), a serverless, event-driven compute service, and [AWS AppSync](#), which connects apps to data and events with secure, serverless, and high-performing GraphQL and Pub/Sub APIs. The choice to use AWS was partially driven to be more sustainable. Because serverless technology can scale dynamically to meet demand for cloud resources and avoid overprovisioned capacity, Sustell uses less energy and has a lower carbon footprint than traditional on-premises infrastructure.

Serverless infrastructure also requires less maintenance and is highly flexible, scaling rapidly to accommodate growth. Thus, dsm-firmenich avoids excessive operational overhead and hardware expenses, thereby maintaining stable costs. “The dsm-firmenich team was keen to create a developer-oriented and high-level stack of AWS services and tools to run Sustell on,” says Andres Koetsier, cloud-native development practice lead at Xebia. “By building on AWS, rather than a traditional infrastructure, we can increase in-house skills and keep a tight control on costs.”

Using Sustell, animal farmers can simulate scenarios to see how changes in their feed and farming practices, such as the type and source of feed ingredients, can reduce environmental impacts. According to a white paper from the [World Wildlife Fund](#), animal feed production and manufacturing in the US and Canada contribute to 20 percent of livestock’s carbon footprint, so relatively small changes can make a large impact. Sustell uses an engine from Blonk Consultants, an expert in the field of agri-footprint measurement, to calculate the footprint of animal production systems and analyze the impact of user decisions.

Sustell democratizes life cycle assessment (LCA), a comprehensive evaluation of the environmental impacts of a product or service throughout its entire life span. Sustell uses internationally certified LCA calculation methodologies (normally reserved for LCA experts), along with real feed, farm, and processing data, to provide accurate analysis of complex and highly nuanced farming systems. Instead of relying on specialists, businesses in the animal protein value chain can take full ownership of their LCA needs and scale accordingly.

LCA data not only benefits animal farmers but also helps banks understand the sustainability and climate-related risks they are taking with certain businesses. Additionally, environmental, social, and governance investors can use LCA data to understand how farms are run and whether they are subject to environmental risks. Sustainability-linked bonds and loans require regular, systematic measur

environmental footprints across various farms and dimensions. And with Sustell, farming businesses can comply with these demands.

Advancing Analytic Capabilities to Support Animal Farmers

As Sustell continues to grow, dsm-firmenich is committed to refining its offerings. The solution's modular, serverless design helps integrate new features; the company has added five new animal modules in 3 months.

Sustell can present very complex data in a digestible manner, a trait that has potential applications for other industries and other business areas within dsm-firmenich. With early adopters providing vast amounts of data, Sustell's analytic and modeling capabilities will keep improving.

dsm-firmenich plans to explore how artificial intelligence could help spot trends in similar farming systems and generate user recommendations. As the industry continues to evolve, animal farmers will continue to trust Sustell to provide a holistic, transparent, and automated digital journey.

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